

HanGuk Shin

Product Engineer · Mobile, Media & Distributed Systems

I build complex products across mobile, media, and backend systems, designing them to converge toward the correct state even when operations fail.

hanguk0726@gmail.com · <https://github.com/hanguk0726> · <https://medium.com/@hangukshin> · Based in Seoul

Core Capabilities

Product Engineering 0 → 1 product building · Structuring requirements · App / web / back-office / backend integration · Shipping and operating · Automating a small team's repetitive work

Mobile Android / Kotlin · Jetpack Compose · React Native / Expo · Flutter · Camera · background tasks · App distribution and OTA

Media & Realtime OpenGL ES · EGL · MediaCodec · Video decode / render / encode · WebRTC · STT / TTS · Real-time audio processing

Backend & Distributed Systems Go · TypeScript · Python · PostgreSQL · MQTT · LoRa · Kafka · ClickHouse · State machines · idempotency · Retry · timeout · recovery · observability

Web & Visualization React · Next.js · WebGL · Pannellum · Back-office · Data visualization

Featured Projects

REVITA

2025.10 – Present

A computer-vision IoT project where I built the mobile apps, public web, back-office, and field-device control backend from scratch.

Role Product engineer — owned the apps (Android/iOS), web, back-office, and field-control backend end to end.

CONTRIBUTION

- Built the product end to end on React Native, Next.js, Python, and PostgreSQL
- Modeled valve control over MQTT (QoS 2) and LoRa as a state machine, keeping state honestly pending until an ACK/NOTIFY arrives
- Added recovery via timeouts, a safety sweep, and per-minute monitoring that nudges stalled sessions back toward correctness
- Make-Before-Break multi-zone control to avoid deadheading a shared pump, with idempotent CAS-based pump start
- Switched panorama generation from image stitching to video-based scanning with OpenCV
- Automated back-office repetitive work such as GeoTIFF overlays and content into shared team workflows
- Built tiered Sentry observability, reproduced issues on real devices, and added regression guards

OUTCOME

- **15 min+ → ~2 min** Panorama generation time
- State-drift recovery — A recovery path that structurally converges even when device and server state drift.
- Repetitive work removed — Individual-dependent repetitive work turned into shared team workflows.

React Native · Expo · TypeScript · Next.js · OpenCV · Pannellum · MQTT · LoRa · PostgreSQL · Airflow · Python

Android Video Editor

2025.02 – 2025.06

An Android multi-source video editor and encoding pipeline wiring MediaCodec, OpenGL ES, and EGL.

Role Android developer (project contract) — designed and built the video pipeline.

CONTRIBUTION

- Designed a Decoder → SurfaceTexture → OpenGL → Encoder pipeline
- Frame processing with OES textures, FBOs, and shaders
- Passed OpenGL output frames straight to the encoder through an EGLSurface
- Timeline control over multiple video and text sources
- Encoded PCM to AAC and muxed it into the final video

OUTCOME

- Removed intermediate YUV conversion — Fed OpenGL output frames to the encoder via EGLSurface, eliminating the conversion step.
- Multi-source composition and timeline editing

Android · Kotlin · OpenGL ES · EGL · MediaCodec · MediaMuxer

Phylaxis Proxy

2023.10 – 2024.01

Built rate limiting, cache, security filtering, and the log/stats pipeline in a Go reverse proxy.

Role Backend Go developer — owned rate limiting, cache, SEO, and the log/stats work.

CONTRIBUTION

- Rate limiting and an LRU RAM/disk cache with TTL management
- Parsed request headers/query/body and analyzed input to defend against SQLi and XSS
- Log and statistics processing with Kafka and ClickHouse
- HAProxy integration (certs, fingerprint, load balancing) and Docker packaging and deploy

OUTCOME

- Resource-aware request path — Designed the multi-stage flow of filtered and processed requests with resource usage in mind.
- Performance, security, and operations together

Go · Aerospike · ClickHouse · Kafka · HAProxy · Docker

Togispace		2022.01 – 2023.02
Built a commercial Android app from scratch and operated it for 14 months through repeated product pivots.		
Role Android developer — sole owner of the client through build and operations.		
CONTRIBUTION		
• Built the app on Kotlin and Jetpack Compose • Clean Architecture and a multi-module structure with event-driven, reactive programming • Auth, camera, PDF, WebView, i18n, and network/image caching • Complex widgets (calendar, cropper, swipe, drag-and-drop) and app state management • Collaborated with other roles across the full product lifecycle		
OUTCOME		
• 14 months operating a B2C app • Structure resilient to repeated pivots — Kept the structure intact through several major pivots by continually improving state management.		
Kotlin · Jetpack Compose		
VocalBridge		2025.08 – 2025.08
A prototype linking real-time voice through STT, translation, and TTS over React Native and WebRTC.		
Role Personal project — built the app and the full pipeline.		
CONTRIBUTION		
• Built the UI with React Native and Expo • Streamed audio over WebRTC • Assembled an STT → translation → TTS pipeline (Korean, English, Japanese) • Processed voice-waveform animation data natively		
OUTCOME		
• Three-language real-time voice translation prototype		
React Native · Expo · TypeScript · WebRTC · FastAPI · Supabase		
Career		
CAREER		
REVITA · Product Engineer		2025.10 – Present
Yongsan, Seoul		
Building the mobile apps, web, back-office, and field-device control backend.		
• React Native Android & iOS apps • Next.js web and back-office • MQTT/LoRa-based IoT control • Observability and operations automation		
Freelance · Android Developer		2025.02 – 2025.06
Project contract		
Built an OpenGL/MediaCodec Android video editor.		
• Designed and built the decode/compose/encode pipeline • Passed render frames to the encoder via EGLSurface		
Phylaxis · Backend Go Developer		2023.10 – 2024.01
Jung-gu, Seoul		
Built cache, rate limit, log, and security features in a Go reverse proxy.		
• Rate limit and LRU cache / TTL management • Input analysis for SQLi/XSS defense • Kafka/ClickHouse logs and statistics		
Togispace · Android Developer		2022.01 – 2023.02
Yeouido, Seoul		
Sole developer and operator of a commercial B2C app on Kotlin/Jetpack Compose.		
• Built a Clean Architecture, multi-module structure • Auth, camera, PDF, WebView, and i18n • Improved app state management through repeated pivots		
Snaptag · Android & Backend Developer		2021.03 – 2021.06
Pangyo, Gyeonggi		
Android camera frame processing and JNI/C++ module integration.		
• Optimized from camera capture to information extraction • Passed camera frames to a C++ module via JNI		
BOOTCAMP		
Cloud — Korea Radio Promotion Association · Cloud training program		2020.05 – 2020.10
Seocho-gu, Seoul		

EDUCATION

Project Archive

- **Avatar Vision** — A video-diary app in Flutter and Rust without ffmpeg — decoding webcam/mic input and muxing to MP4.
- **CrowdFunding** — Crowdfunding on a smart contract — campaign creation, funding, and claim, syncing frontend CRUD with on-chain transactions.
- **Realtime Chat** — Multi-channel real-time messaging with a websocket pool, read receipts, and magic-link login.
- **Plantimer** — A productivity app that schedules tasks as a progress-bar timer, continuing via notifications when closed.
- **Strawberry Days** — A fan site for singer Younha — aggregated social updates, album/track browsing, and community.
- **Snaptag** — Scanning invisible QR codes with the Android camera — passing frames to a C++ module via JNI and optimizing capture-to-extraction.
- **Openshy4J** — An OpenStack web controller to create and manage instances via Spring and openstack4j.
- **Fivestar Recipe** — A cooking-recipe site with bookmarks, search, and ranking (Spring MVC, Vue.js).
- **Blue Marble** — My first project after learning Java — Monopoly-style board-game logic in the terminal.